

Step 2: Critical Thinking Questions

1. The Barrier: When using the mobile hotspot, what is the difference between your "Internal IP" and your "Public IP"? Why does the VPN client need the Public IP? An "Internal IP" (or Private IP) is assigned by your local home router to devices inside your network and cannot be routed across the internet. A "Public IP" is assigned to your modem or router by your Internet Service Provider and is globally visible. When your client is on a mobile hotspot, it is completely disconnected from your local network. Therefore, the VPN client must connect to the Public IP so the traffic can successfully route across the internet to reach your home router.

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2. Persistence: If you are banned by Fail2Ban because you forgot your SSH key, how would you get back into the server? Because Fail2Ban enforces bans based on IP addresses for a specified duration, you have a few ways to regain access:

- **Wait for the ban to expire:** Your Fail2Ban configuration enforces a bantime of 1 hour, so you could simply wait it out.

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- **Change your IP address:** Disconnect from the mobile hotspot and connect to a different network (such as another Wi-Fi network) to obtain a new, unbanned IP address.
- **Physical Access:** Access the physical Gateway machine directly with a keyboard and monitor, log in, and manually unban the IP address using the Fail2Ban client.

3. Stealth: Try to ping your Public IP from the hotspot client. Does it respond? Why is this beneficial? It does not respond. This occurs because the gateway's iptables firewall was specifically configured to drop ICMP echo requests (pings) arriving on the public eth0 interface. This is beneficial because it keeps the gateway "invisible" to public scanners and automated bots. By not acknowledging the ping, attackers are less likely to discover that a server exists at that IP address.

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Reflection

1. Hole Punching: Explain how the Gateway was able to connect to the Client without you touching the router settings. By utilizing an overlay engine like Tailscale, the connection bypasses the need for manual port forwarding. Both the Gateway and the Client initiate an *outbound* connection to an external coordination server. Firewalls and routers naturally allow outbound traffic and keep a temporary path open for the return response. The coordination server utilizes these newly opened paths to facilitate a direct, "hole-punched" connection between the two peers without requiring router configuration changes.

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2. Traffic Analysis: Why is it more secure to use an Overlay Network than to open Port 22 on a public-facing router? Opening Port 22 directly on a public router exposes your SSH management interface to the entire internet, which invites continuous automated bot scans and brute-force attacks. Using an Overlay Network obfuscates this by requiring cryptographic authentication before the devices can even communicate. By moving SSH to an overlay network and a high port like 2222, you separate management traffic from tunnel traffic, completely hiding your control plane from unauthorized internet users.

3. The Hotspot Factor: How did the latency change when you switched from the local lab to the Mobile Hotspot? Did the VPN tunnel remain stable? Latency (ping time) significantly increases when switching to a mobile hotspot because the data must travel from your laptop to a cell tower, route through the cellular provider's infrastructure across the public internet, and finally arrive at your home network, rather than just passing through a local switch. Despite this latency increase and the complete change in your external IP address, the VPN tunnel remains stable. Because the underlying protocol (WireGuard) is stateless, it relies on cryptographic keys rather than IP addresses to maintain the connection, allowing for seamless roaming.

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